**F78PA Probability & Statistics A**

**Tutorial 2 – Solutions**

1. (a) The set of outcomes is given by $\Omega = \{1, 2, 3, 4, 5, 6\}$ and we can define $X(\omega) = \omega$. Then $\text{Im}(X) = X(\Omega) = \Omega = \{1, 2, 3, 4, 5, 6\}$ which is a finite set and hence countable. Therefore $X$ is a discrete random variable with $\text{Im}(X) = \{1, 2, 3, 4, 5, 6\}$.

   (b) The set of outcomes is given by $\Omega = [0, 1]$ and $X(\omega) = \omega$ so $\text{Im}(X) = X(\Omega) = \Omega = [0, 1]$. Since the interval $[0, 1]$ is uncountable we have that $X$ is not a discrete random variable.

   (c) Let $\Omega = \{\omega = (\omega_i)_{i=1}^{\infty} : \omega_i \in \{H, T\}\}$ be the set of sequences of Heads, $H$, and Tails, $T$. Then
   $$X(\omega) = \inf\{i \in \{1, 2, 3, \ldots\} : \omega_i = H\}.$$
   In this case the image of $X$ is clearly contained within $\{1, 2, 3, \ldots\}$, and for each fixed $n \in \{1, 2, 3, \ldots\}$ we can define a sequence $\omega \in \Omega$ by $\omega_n = H$ and $\omega_i = T$ for $i \neq n$. Therefore $\text{Im}(X) = \{1, 2, 3, \ldots\}$ and $X$ is a discrete random variable.

   (d) Let $\Omega = \{\omega \in R^2 : ||\omega|| \leq 1\}$ be the point which the dart hits (here we are assuming the dart always hits the board, otherwise we can take $\Omega = \mathbb{R}^2$), considering $(0, 0)$ to be the centre of the board. Then we set $X(\omega) = ||\omega||$ to be the distance between $\omega$ and the centre of the board. Then the image of $X$ is given by $\text{Im}(X) = [0, 1]$ and $X$ is not a discrete random variable.

   (e) Let $\Omega = [0, 10]$ and $X(\omega) = \lfloor \omega \rfloor$ then the image of $X$ is $\text{Im}(X) = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. As the image is a finite set it is countable and hence $X$ is a discrete random variable.

   (f) Let $\Omega = \{\omega \in R^2 : ||\omega|| \leq 1\} \cup \{\text{miss}\}$ be the point which the dart hits (for simplicity we have scaled the board to have radius 1). Let $X(\omega)$ to be the value of the point where the dart hits. Then the image of $X$ is given by
   $$\text{Im}(X) = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}.$$
   Since this set is finite it is countable and hence $X$ is a discrete random variable.

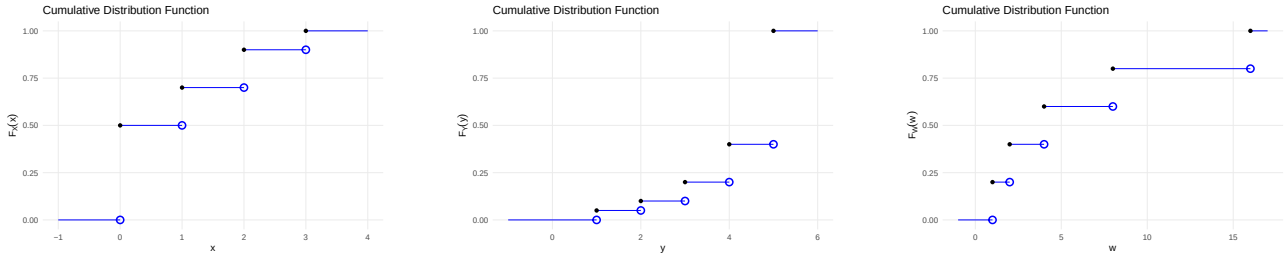
   (g) Let $\Omega = \{1, 2, 3, \ldots\}$ be the set of possible labels drawn, then set $X(\omega) = \omega$ for all $\omega \in \Omega$. Then the image of $X$ is $\Omega$ which is countable and hence $X$ is a discrete random variable.

2. (a) The cumulative distribution function, $F_X$, of $X$ is given by
   $$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.5, & 0 \leq x < 1, \\ 0.7, & 1 \leq x < 2, \\ 0.9, & 2 \leq x < 3, \\ 1, & 3 \leq x. \end{cases}$$

   (b) The cumulative distribution function, $F_Y$, of $Y$ is given by
   $$F_Y(y) = \begin{cases} 0, & y < 1, \\ 0.05, & 1 \leq y < 2, \\ 0.1, & 2 \leq y < 3, \\ 0.2, & 3 \leq y < 4, \\ 0.4, & 4 \leq y < 5, \\ 1, & y \geq 5. \end{cases}$$

(c) The cumulative distribution function $F_W$ of $W$ is given by

$$F_W(w) = \begin{cases} 0, & w < 1, \\ 0.2, & 1 \leq w < 2 \\ 0.4, & 2 \leq w < 4 \\ 0.6, & 4 \leq w < 8 \\ 0.8, & 8 \leq w < 16 \\ 1, & w \geq 16. \end{cases}$$




(a) CDF of X (b) CDF of Y (c) CDF of W

Figure 1: Comparison of cumulative distribution functions for X, Y, and W

3. Probability mass function satisfies $\sum_x p_X(x) = 1$, so

$$\begin{aligned} (a+b) + (2a+b) + (3a+b) + (4a+b) + (5a+b) &= 1 \\ \Rightarrow \quad 15a + 5b &= 1 \\ \Rightarrow \quad 3a + b &= \frac{1}{5} \end{aligned}$$

Also, we are told that $\mathbb{P}(X \geq 3) = \frac{1}{2}$ so

$$\frac{1}{2} = \mathbb{P}(X \geq 3) = p_X(3) + p_X(4) = (4a+b) + (5a+b) = 9a + 2b.$$

So we need to solve the simultaneous equations

$$\begin{aligned} 3a + b &= \frac{1}{5} \\ 9a + 2b &= \frac{1}{2} \end{aligned}$$

From the first equation we have $b = \frac{1}{5} - 3a$ substituting this into the second equation gives

$$9a + 2\left(\frac{1}{5} - 3a\right) = \frac{1}{2} \Rightarrow 3a + \frac{2}{5} = \frac{1}{2} \Rightarrow 3a = \frac{1}{10}$$

Therefore $a = \frac{1}{30}$ and $b = \frac{1}{5} - \frac{3}{30} = \frac{1}{10}$.

4. (a) With $X \sim \text{Bin}(5, 0.2)$, then $\mathbb{P}(X = 3) = \binom{5}{3} 0.2^3 0.8^2 = 0.0512$

   (b) With $X \sim \text{Bin}(10, 0.3)$, then $\mathbb{P}(X \leq 2) = \binom{10}{0} 0.3^0 0.7^{10} + \binom{10}{1} 0.3^1 0.7^9 + \binom{10}{2} 0.3^2 0.7^8 \approx 0.3828$

   (c) With $X \sim \text{Bin}(15, 0.05)$, then $\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X \leq 1) = 1 - \binom{15}{0} 0.05^0 0.95^{15} - \binom{15}{1} 0.05^1 0.95^{14} \approx 0.1710$

   (d) With $X \sim \text{Bin}(8, 0.7)$, then $\mathbb{P}(X > 6) = \mathbb{P}(X \geq 7) = \binom{8}{7} 0.7^7 0.3^1 + \binom{8}{8} 0.7^8 0.3^0 \approx 0.2553$

   (e) With $X \sim \text{Bin}(10, 0.6)$, then $\mathbb{P}(X \geq 8.3) = \mathbb{P}(X \geq 9) = \binom{10}{9} 0.6^9 0.4^1 + \binom{10}{10} 0.6^{10} 0.4^0 \approx 0.0464$

5. Denoting by $X$ the number of sales, then $X \sim \text{Bin}(12, 0.08)$.

   i. $\mathbb{P}(X \leq 2) = \binom{12}{0}0.08^0 0.92^{12} + \binom{12}{1}0.08^1 0.92^{11} + \binom{12}{2}0.08^2 0.92^{10} \approx 0.9348$

   ii. $\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X \leq 1) = 1 - \binom{12}{0}0.08^0 0.92^{12} - \binom{12}{1}0.08^1 0.92^{11} \approx 0.2487$

6. Denoting by $X$ the number of packets containing traces of nuts, then $X \sim \text{Bin}(20, 0.06)$.

   i. $\mathbb{P}(X = 0) = \binom{20}{0}0.06^0 0.94^{20} \approx 0.2901$

   ii. $\mathbb{P}(X > 2) = 1 - \mathbb{P}(X \leq 2) = 1 - \binom{20}{0}0.06^0 0.94^{20} - \binom{20}{1}0.06^1 0.94^{19} - \binom{20}{2}0.06^2 0.94^{18} \approx 0.1150$

7. Let $X$ denote the number of components that fail, then $X \sim \text{Bin}(3, 0.05)$.

$$\mathbb{P}(X \geq 2) = \mathbb{P}(X = 2) + \mathbb{P}(X = 3) = \binom{3}{2}(0.05)^2(0.95)^1 + \binom{3}{3}(0.05)^3(0.95)^0 = 0.00725.$$

   So, the probability that the system will overheat is 0.00725.

8. Let $X$ be the number of people who have received the vaccine and will contract influenza A. Then, $X \sim \text{Bin}(n, 0.10)$, where $n$ is the number of people who have received the influenza A vaccine, $p = 1 - 0.90 = 0.10$. We need to find $n$ such that

$$\mathbb{P}(X \geq 1) \geq 0.95.$$

   Note

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - \binom{n}{0}0.10^0\, 0.9^n.$$

   Therefore it remains to find $n$ such that $0.9^n \leq 0.05$, which gives the condition

$$n \geq \frac{\log(0.05)}{\log(0.90)} \approx 28.4331.$$

   Thus, to satisfy the conditions of this probability, a minimum of 29 vaccinated individuals need to be exposed to influenza A.

9. For $X \sim \text{Bin}(n, p)$ with $0 \leq p < 1$, for $r = 0, 1, 2, \ldots, n-1$,

$$\begin{aligned}
\frac{\mathbb{P}(X = r+1)}{\mathbb{P}(X = r)} &= \frac{\binom{n}{r+1}p^{r+1}(1-p)^{n-(r+1)}}{\binom{n}{r}p^r(1-p)^{n-r}} \\
&= \frac{\frac{n!}{(r+1)!\,(n-(r+1))!}}{\frac{n!}{r!\,(n-r)!}}\,\frac{p}{(1-p)} \\
&= \frac{r!\,(n-r)!}{(r+1)!\,(n-(r+1))!}\,\frac{p}{1-p} \\
&= \left(\frac{n-r}{r+1}\right)\left(\frac{p}{1-p}\right)
\end{aligned}$$

   With $x_0 = (n+1)p - 1$, then

   - For $r \leq x_0$,

$$\begin{aligned}
\frac{\mathbb{P}(X = r+1)}{\mathbb{P}(X = r)} &= \left(\frac{n-r}{r+1}\right)\left(\frac{p}{1-p}\right) \\
&\geq \left(\frac{n-x_0}{x_0+1}\right)\left(\frac{p}{1-p}\right) \\
&= \left(\frac{n-(n+1)p+1}{(n+1)p}\right)\left(\frac{p}{1-p}\right) = \left(\frac{(n+1)(1-p)}{(n+1)p}\right)\left(\frac{p}{1-p}\right) = 1,
\end{aligned}$$

     so for $0 \leq r \leq x_0$, we have $\mathbb{P}(X = r+1) \geq \mathbb{P}(X = r)$, as required.

- For $r \geq x_0$,

$$\begin{aligned}\frac{\mathbb{P}(X=r+1)}{\mathbb{P}(X=r)} &= \left(\frac{n-r}{r+1}\right)\left(\frac{p}{1-p}\right) \\ &\leq \left(\frac{n-x_0}{x_0+1}\right)\left(\frac{p}{1-p}\right) = 1 \text{ (same algebra as above),}\end{aligned}$$

so for $x_0 \leq r \leq n-1$, we have $\mathbb{P}(X=r+1) \leq \mathbb{P}(X=r)$, as required.

The two inequalities above show that

$$\mathbb{P}\left(X=\lfloor x_0 \rfloor + 1\right) \geq \mathbb{P}\left(X=\lfloor x_0 \rfloor\right) \geq \mathbb{P}\left(X=\lfloor x_0 \rfloor - 1\right) \geq \ldots \geq \mathbb{P}(X=1) \geq \mathbb{P}(X=0)$$

and

$$\mathbb{P}(X=n) \leq \mathbb{P}(X=n-1) \leq \ldots \leq \mathbb{P}\left(X=\lfloor x_0 \rfloor + 2\right) \leq \mathbb{P}\left(X=\lfloor x_0 \rfloor + 1\right)$$

So $\mathbb{P}(X=r)$ is non-decreasing for $r = 0, 1, \ldots, \lfloor x_0 \rfloor + 1$, and then non-increasing for $r = \lfloor x_0 \rfloor + 1, \ldots, n-1, n$, achieving its maximal value at $r = \lfloor x_0 \rfloor + 1 = \lfloor (n+1)p - 1 \rfloor + 1 = \lfloor (n+1)p \rfloor$. The Binomial distribution $\text{Bin}(n,p)$ has a mode at $\lfloor (n+1)p \rfloor$. This mode may not be unique; if $(n+1)p$ is an integer, then $(n+1)p - 1$ and $(n+1)p$ are both modes.

## Exam Style Question

Define the events

$$\begin{aligned}A &= \text{component produced in Aberdeen,} \\ B &= \text{component produced in Belfast,} \\ C &= \text{component produced in Cardiff,} \\ D &= \text{component defective.}\end{aligned}$$

Then we are told that

$$\begin{array}{lll}\mathbb{P}(A) = 0.3, & \mathbb{P}(B) = 0.3, & \mathbb{P}(C) = 0.4, \\ \mathbb{P}(D|A) = 0.01, & \mathbb{P}(D|B) = 0.03, & \mathbb{P}(D|C) = 0.02.\end{array}$$

(a) Using the law of Total Probability

$$\begin{aligned}\mathbb{P}(D) &= \mathbb{P}(A)\mathbb{P}(D|A) + \mathbb{P}(B)\mathbb{P}(D|B) + \mathbb{P}(C)\mathbb{P}(D|C) \\ &= 0.3 \times 0.01 + 0.3 \times 0.03 + 0.4 \times 0.02 \\ &= 0.02.\end{aligned}$$

[2 marks]

(b) Using Bayes' Theorem

$$\mathbb{P}(A|D) = \frac{\mathbb{P}(A)\mathbb{P}(D|A)}{\mathbb{P}(D)} = \frac{0.3 \times 0.01}{0.02} = \frac{3}{20}.$$

[2 marks]

(c) i) To justify treating $X$ as binomially distributed we need to assume:

- Each trial is independent, i.e. that whether or not a particular component is defective is independent of whether or not any other components are defective;
- Each trial has equal probability of being successful, i.e. each component has the same probability of being defective.

[2 marks]

ii) Under the above assumptions, from part (a), $X \sim \text{Bin}(20, 0.02)$. Therefore

$$\begin{aligned}
\mathbb{P}(X > 2) &= 1 - \mathbb{P}(X \leq 2) \\
&= 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1) - \mathbb{P}(X = 2) \\
&= 1 - \binom{20}{0} 0.02^0 0.98^{20} - \binom{20}{1} 0.02^1 0.98^{19} - \binom{20}{2} 0.02^2 0.98^{18} \\
&= 1 - 0.667608 - 0.272493 - 0.052830 \\
&= 0.0070.
\end{aligned}$$

[3 marks]

**[Total 9 marks]**